Humanitarian Logistics Hub Network Design Under Uncertainty: A Multi-Objective Model and Priority-Based Genetic Algorithm for Central Coastal Vietnam

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Abstract. Conventional disaster relief models often assume uninterrupted road networks, making them inadequate for Central Vietnam, where frequent floods severely disrupt infrastructure. We formulate the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP), a two-stage stochastic program that jointly minimizes expected logistics cost and expected maximum deprivation cost over a set of flood scenarios. The model explicitly accounts for broken inter-hub links, multi-modal transport (land, water, and air), pre-positioned inventory, as well as adaptive lateral transshipments and reactive hub activations to recover from severe localized deficits. Deprivation cost is modeled as an exponential penalty on victim waiting time, and Daganzo's continuous approximation is embedded for tractable last-mile routing cost estimation. To solve MO-IHLNDP, we propose PB-NSGA (Priority-Based Nondominated Sorting Genetic Algorithm), which confines the chromosome to first-stage hub decisions and reconstructs all scenario-dependent second-stage variables through a structured priority-based decoder, reducing search-space complexity from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}| + |\mathcal{I}|)$. On CV-Small, PB-NSGA achieves HV = 0.925 in 1.1 s, closely approaching the MILP reference front (HV = 0.997, 4.9 s) and outperforming VNS-TS (HV = 0.777, 60 s) and GWO-HD (HV = 0.780, 0.6 s). On the full-scale CV-Large instance, PB-NSGA defines the combined reference front (HV = 1.000 ± 0.006) in under 10 s – versus 180 s for VNS-TS (HV = 0.793). A case study across four flood-prone provinces in Central Vietnam yields a decision-support framework: the Quang Ngai Port Hub and Phuoc Son Helipad are unconditional network anchors (100% Pareto-selection frequency), followed by the Huong Viet Depot (82.8%) and Da Nang Airport Hub (77.3%), while the Tam Ky Logistics Hub (67.2%) is identified as a resilience gap requiring contingency planning.

Keywords: Humanitarian logistics · Hub location · Multi-objective optimization · Stochastic programming · Disaster relief · Evolutionary algorithm

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1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [1]. As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [2].

Humanitarian operations during this period faced significant challenges. Most critically, national logistics infrastructure suffered severe disruption: the North-South Railway was paralyzed by track erosion and submersion, forcing the cancellation of 170 trains, while key arteries such as Vietnam National Route 1 were severed by landslides, isolating communities for days [2]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed – the physical network becomes "incomplete", characterized by broken links and isolated nodes. Compounding this, resource scarcity arising from poor planning and uncertain supply and demand further strains relief efforts. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing [3].

1.2 Related Works

Humanitarian logistics and facility location. Humanitarian logistics differs fundamentally from its commercial counterpart in that speed and equity dominate over cost minimisation [4]. A comprehensive review of humanitarian facility location under uncertainty [5] identifies stochastic programming and robust optimisation as the dominant paradigms, while Boonmee et al. [6] document the growing need for models that jointly capture disruption, uncertainty, and equity. To account for social welfare, Holguín-Veras et al. [7] introduced *deprivation cost* – an exponential penalty on victim waiting time.

Hub location and incomplete networks. Hub-and-spoke structures consolidate flows efficiently and have been studied extensively since Campbell's seminal formulations [8]. For disaster settings, the *incomplete* hub network – in which infrastructure damage severs inter-hub arcs – is a more realistic topology than the classical fully-connected graph [9]. Sangsawang and Chanta [10] proposed a pioneering flood-specific hub location model, a single-objective deterministic formulation for Thailand, establishing that hub-and-spoke structures yield measurable gains in relief distribution efficiency. Multi-modal hub networks further address the heterogeneous transport modes (road, waterway, air) that characterise disaster response [11].

Stochastic and robust optimisation for relief networks. The two-stage stochastic programming framework naturally separates pre-disaster strategic decisions from post-disaster recourse [12]. Under demand and supply uncertainty, robust formulations [13] and prepositioning models have been shown to outperform deterministic baselines. Disruption of transportation infrastructure compounds this uncertainty by rendering pre-planned routes infeasible [14], a condition endemic to Central Vietnam's flood disasters.

Multi-objective evolutionary algorithms. NSGA-II [15] is the standard framework for multi-objective combinatorial problems in humanitarian logistics, owing to its parameter-free Pareto ranking and elitist selection. Zhou et al. [16] applied a multi-objective evolutionary algorithm to multi-period dynamic emergency resource scheduling under uncertain road networks, demonstrating the effectiveness of population-based search when scenario-dependent decisions are coupled to shared strategic variables. Li et al. [11] designed a bi-objective multimodal hub-and-spoke network for emergency relief using a meta-heuristic approach, showing that problem-specific customized operators are essential for maintaining feasibility across complex network topologies.

Research gap and contributions. To the best of our knowledge, existing literature rarely simultaneously addresses incomplete hub network design, two-stage stochastic programming, multi-objective humanitarian objectives (logistics cost and deprivation cost), and a flood-disaster case study. Sangsawang and Chanta [10] is the closest precedent, yet it is deterministic and single-objective. This paper closes the gap by formulating the MO-IHLNDP and solving it via our proposed PB-NSGA, a priority-based [17] evolutionary algorithm, as this encoding strategy is proven to be particularly effective for handling the complex decision variables of two-stage stochastic programming.

2 Problem Description and Mathematical Model

We model the network design problem as a capacitated, single-allocation MO-IHLNDP. The network comprises **origins** (supply sources), **demands** (victim groups), and **hubs** of two types: planned (pre-disaster) and reactive (post-disaster, dynamically opened).

From a practical standpoint, DRND involves two critical uncertainty factors. First, infrastructure disruption – damaged roads and buildings – can disable pre-programmed plans, necessitating a dynamic solution [13]. Second, the exact locations of demands, the total number of victims, and the availability of origins are all highly uncertain during rescue operations [12]. To account for these, we employ a two-stage stochastic programming approach across multiple scenarios, each assigning a specific risk factor per area, and assume that Search and Rescue operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy uncertain demands under extreme scenarios.

The model integrates the deprivation cost [7] for social equity, and Daganzo's continuous approximation (CA) [18] to tractably estimate last-mile rescue times.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$	Set of candidate hubs (k, h), demand nodes (i), and origins (j)
$\mathcal{S}, \mathcal{M}$	Set of disaster scenarios (s) and transportation modes (m)
$\mathcal{V}$	Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k	Fixed cost to establish hub k of the first type (before the disaster)
c_k, κ_k	Unit holding cost and capacity limit for inventory at hub k
α, χ, M	Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
C_{uvm}, τ_{uvm}	Unit transportation cost and estimated travel time from node u to v via mode m
C_m, Q_m	Unit local routing cost and transportation capacity for vehicle mode m
γ	Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F^a_{ks}	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
a_{uvms}	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u, and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i, and relief items provided at origin j in scenario s
c_{jks}	Pre-computed cheapest transport cost from origin j to hub k in scenario s: $c_{jks} = \min_{m \in \mathcal{M} \mid a_{jkms}=1} C_{jkm}$, ($+\infty$ if no available mode)
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuity factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s, 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s, 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s, 0 otherwise
q_k	Amount of relief items inventoried at hub k
f_{khms}	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$\begin{aligned}
\text{Min } Z_1 = & \underbrace{\sum_{k\in\mathcal{H}} F_k x_k + \sum_{k\in\mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
& + \sum_{s\in\mathcal{S}} \pi_s \Bigg[\underbrace{\sum_{k\in\mathcal{H}} F^a_{ks} y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j\in\mathcal{J}} \sum_{k\in\mathcal{H}} c_{jks} \cdot O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \\
& + \underbrace{\sum_{k\in\mathcal{H}} \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k\in\mathcal{H}} \sum_{i\in\mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \Bigg]
\end{aligned} \tag{1}$$

Objective Z_1 minimizes total expected logistics cost. Using the previously mentioned CA [18], demands are aggregated on a grid basis to pre-compute the last-mile routing cost Θ_{kis}. The first term is line-haul cost from hub k to demand i, and the second term is local detour cost inside area i:

$$\Theta_{kis} = \min_{m\in\mathcal{M}|a_{ikms}=1} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s\in\mathcal{S}} \pi_s \left(\max_{i\in\mathcal{I}} \left[D_{is} \cdot \left(e^{\lambda_{is} \cdot \Omega_{is}} - 1 \right) \right] \right) \tag{2}$$

$$\Omega_{is} = \sum_{k\in\mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m\in\mathcal{M}|a_{ikms}=1} \tau_{kim} \right)$$

Objective Z_2 minimizes the expected value of maximum deprivation cost [7]. The exponential function of waiting time Ω_{is} heavily penalizes the model if it lets a demand node wait slightly longer than the rest, forcing the model to distribute resources evenly. The penalty function is adjusted by $\lambda_{is} = \lambda_0(1 + r_{is})$ so that highly vulnerable populations are prioritized.

To linearize Z_2, we exploit the single-allocation property (8). Any demand i is allocated to a single hub k. Thus, the deprivation cost C^{dep}_{iks} is pre-computed:

$$C^{\text{dep}}_{iks} = D_{is} \cdot \left(e^{\lambda_{is} \cdot \left(\tau_{ks} + 2 \cdot \min_{m\in\mathcal{M}|a_{ikms}=1} \tau_{kim} \right)} - 1 \right) \tag{3}$$

By introducing a continuous auxiliary variable $W_s \geq 0$ and constraint (4) to capture the worst-case deprivation cost among all demand nodes in a given scenario s, we obtain the linearized program:

$$\begin{aligned} \text{Minimize} \quad & Z_1, Z_2^{\text{linear}} = \sum_{s\in\mathcal{S}} \pi_s \cdot W_s \\ \text{subject to} \quad & \text{Constraints (4)} - (18) \end{aligned}$$

$$W_s \geq C_{iks}^{\text{dep}} \cdot z_{iks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{4}$$

2.5 Constraints

The original, complete MO-IHLNDP model is formally stated as follows:

$$\begin{aligned} \text{Minimize} \quad & Z_1, Z_2 \\ \text{subject to} \quad & \text{Constraints (5)} - (18) \end{aligned}$$

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{5}$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \tag{6}$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{7}$$

$$\sum_{k\in\mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \tag{8}$$

$$\sum_{k\in\mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \tag{9}$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{10}$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{11}$$

$$z_{iks} \leq \sum_{m\in\mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{12}$$

$$z_{jks} \leq \sum_{m\in\mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{13}$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H},\ m \in \mathcal{M},\ s \in \mathcal{S} \tag{14}$$

$$\begin{aligned} \sum_{i\in\mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{khms} \leq q_k &+ \sum_{j\in\mathcal{J}} O_{js} z_{jks} \\ &+ \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \end{aligned} \tag{15}$$

$$q_k + \sum_{j\in\mathcal{J}} O_{js} z_{jks} + \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{16}$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \tag{17}$$

$$q_k, f_{khms}, W_s \geq 0 \quad \forall k, h, m, s \tag{18}$$

Constraints (5) define the hub establishment. For planned hubs, inventory is bounded by physical capacity (6), while reactive hubs are required to be placed within safe zones (7). Constraints (8)-(13) enforce the single-allocation property to active hubs via available links. Constraint (14) restricts flow to be trans-shipped on intact paths. Constraints (15) and (16) maintain flow conservation and capacity limits. Specifically, (15) ensures that total supply, comprising inventory (q_k), external origins (O_{js}), and incoming flows – satisfies both victim demand (converted by γ) and outgoing trans-shipments. Variable domains are defined in (17)-(18).

3 Proposed Algorithm: PB-NSGA

The MO-IHLNDP is NP-hard, as its single-objective, single-scenario relaxation subsumes the uncapacitated facility location problem [19]. We therefore propose **PB-NSGA** (Priority-Based Nondominated Sorting Genetic Algorithm), which embeds the NSGA-II framework [15] with a custom heuristic decoder.

3.1 Chromosome Encoding Scheme

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$ consists of four segments. The binary vector $\mathbf{X} \in \{0,1\}^{|\mathcal{H}|}$ determines planned hub establishment. The continuous vector $\mathbf{R} \in [0,1]^{|\mathcal{H}|}$ defines the inventory fill fraction, setting $q_k = R_k \cdot \kappa_k$. The integer vector $\mathbf{A} \in \{0, \ldots, |\mathcal{H}| - 1\}^{|\mathcal{I}|}$ encodes the preferred anchor hub for each demand i. The continuous vector $\mathbf{W} \in [0,1]^6$ contains six heuristic weights governing the decoder: w_0 and w_3 dictate demand sorting priority by urgency ($\lambda \cdot D$) and isolation; w_1, w_2, w_4 score candidate hubs by travel time, residual inventory, and preference for planned over reactive hubs; w_5 controls conservatism towards opening new reactive hubs.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked once per chromosome per scenario $s \in \mathcal{S}$, reconstructing all second-stage decisions from $\mathbf{C}$ in four steps.

Step 1 – Initialisation. Planned hubs with $X_k = 1$ and $r_{ks} \leq \chi$ are activated with pre-positioned inventory $q_k = R_k \cdot \kappa_k$. If none qualify, the safest hub is forced active.

Step 2 – Demand Prioritisation. Each demand node i receives a composite score prioritising urgency, isolation, and proximity to active hubs. Demands are then processed in descending score order.

Step 3 – Tiered Hub Assignment. Each demand i is assigned to the best reachable active hub, scored by travel time and residual capacity, starting from its anchor π_{A_i}. If all candidates are at capacity (*Pass 2*), the assignment still proceeds with a capacity violation. Only when no reachable active hub exists does the decoder open a new reactive hub as a last resort, incurring its setup cost and incrementing CV.

Step 4 – Supply Routing and Objective Accumulation. Origins supply the most under-supplied active hubs, and surplus is redistributed via discounted

Algorithm 1 Priority-Based Decoder (single scenario s)

Require: Chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$, scenario s, instance data
Ensure: $(Z_1^s,\ Z_2^s,\ \mathrm{CV}^s)$

Activate $\mathcal{H}_{\text{act}} \leftarrow \{k : X_k = 1,\ r_{ks} \le \chi\}$; set $q_k \leftarrow R_k \kappa_k$
if $\mathcal{H}_{\text{act}} = \emptyset$ **then**
  Force-activate $\arg\min_k r_{ks}$ among $\{k : X_k = 1\}$
end if
Score demands by urgency, isolation, and proximity; sort descending
for each demand i (in priority order) **do**
  $K \leftarrow \max(1, \lceil w_5 |\mathcal{H}| \rceil)$; trial order $\leftarrow \pi_{A_i}$
  Pass 1: $k^* \leftarrow \arg\max \mathrm{HubScore}(k)$ over first K active, reachable hubs with residual > 0
  if Pass 1 succeeds **then**
    Assign $i \to k^*$; deduct demand from residual inventory of k^*
  else if any active reachable hub remains **then**
    Pass 2: assign i to first such hub (capacity violation allowed)
  else
    Reactive fallback: open nearest safe inactive hub; add F^a_{ks}; $\mathrm{CV}^s \mathrel{+}= 1$
  end if
end for
Route supply from origins to hubs and redistribute surplus via MCF on the hub sub-network.
Accumulate Z_1^s, Z_2^s
return $(Z_1^s,\ Z_2^s,\ \mathrm{CV}^s)$

inter-hub trans-shipment (α). Expected objectives Z_1 and Z_2 are accumulated across all $s \in \mathcal{S}$. Algorithm 1 formalises these procedures.

With hub activations and demand assignments fixed by Steps 1–3, the supply routing in Step 4 reduces to a **minimum cost flow (MCF)** sub-problem: origins are sources, under-supplied hubs are sinks, and surplus inventory is redistributed via lateral transshipment arcs. Solving MCF exactly per scenario ensures that Z_1 is minimised to near-optimality for any decoded hub configuration, and is a key driver of PB-NSGA's solution quality.

3.3 Genetic Operators

Initialization. The population of N chromosomes is seeded with randomly generated individuals: $\mathbf{X}$ opens between 1 and $\lfloor 0.6|\mathcal{H}| \rfloor$ hubs uniformly at random, $\mathbf{R}$ is drawn from a discrete set of pre-positioning levels, and $\mathbf{A}$, $\mathbf{W}$ are sampled uniformly.

Crossover with probability p_c applies Uniform Crossover to $\mathbf{X}$ and $\mathbf{A}$, and Simulated Binary Crossover (SBX, index η_{SBX}) to $\mathbf{R}$ and $\mathbf{W}$.

Mutation with base rate p_m applies bit-flip to $\mathbf{X}$, random-reset to $\mathbf{A}$, and Polynomial mutation (index η_{poly}) to $\mathbf{R}$ and $\mathbf{W}$. Each gene is mutated independently with probability $p_m/|\text{segment}|$, keeping the expected number of mutations per offspring constant across segments of different lengths.

Survival follows the NSGA-II elitist scheme with constrained dominance [15]: feasible solutions always dominate infeasible ones; among infeasible, lower CV wins. Minimum Hamming distance in $\mathbf{X}$-space serves as a final tiebreaker when crowding distance is tied, preserving diversity in hub configuration.

Diversity. The base mutation rate p_m follows a linear decay schedule from p_m^{high} to p_m^{low} over G generations, balancing early exploration with late exploitation. The adaptive elitist-population strategy [20] is applied to maintain diversity. Further, a counter keeps track of the rank-1 front and conditionally increases tournament size and $\mathbf{W}$-segment mutation rate upon stagnation.

4 Computational Experiments

4.1 Experimental Setup

PB-NSGA is implemented in C++17; dataset construction and post-processing are performed in Python 3, on a workstation with an 11th Gen Intel Core i5-1135G7 @ 2.40 GHz and 16 GB RAM. Algorithm parameters are fixed for all runs: $N = 200$, $G = 300$ (CV-Small) / 500 (CV-Large), $p_c = 0.98$, $p_m^{\text{high}} = 0.40$, $p_m^{\text{low}} = 0.10$ (linear decay), $\eta_{\text{SBX}} = 1.5$, $\eta_{\text{poly}} = 8$. The parameters are chosen via a grid search process. Problem constants are $\chi = 0.7$, $\alpha = 0.6$, $\gamma = 3.0$ kg/person, $\lambda_0 = 0.8$. Each configuration is executed for 20 independent runs, reporting mean $\pm$ std. We use **Hypervolume (HV)** [21] and **IGD$^+$** [22] as performance metrics, computed against the combined non-dominated reference front pooled across all algorithms.

4.2 Dataset Description

Central Vietnam Dataset. We construct two instances modelling four flood-prone provinces: Da Nang, Quang Nam, Thua Thien-Hue, and Quang Ngai. **CV-Small** uses district-level demand nodes ($|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$); **CV-Large** uses commune-level resolution ($|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$). Node coordinates map to real administrative centroids, confirmed logistics facilities, and vital supply origins. Scenario risks and road disruptions are calibrated using a multi-criteria flood vulnerability score [23]. Three transport modes – trucks, motorboats, and helicopters – are considered, with helicopter links capped at 15% of active routing connections per scenario.

Scenario generation via SAA [24]. To improve coverage of plausible flood conditions, scenarios are drawn from a profile bank spanning mild, severe, and extreme flood regimes with varying epicenter count, intensity, road disruption rate, and circuity coefficient. The CV-Large case study further validates the first-stage decisions $(\hat{x}, \hat{q})$ on an independent out-of-sample (OOS) set of $N' = 10$ adversarial scenarios, disjoint from training, to assess generalization under catastrophic conditions.

4.3 Experiment 1: Baseline Comparison

We evaluate PB-NSGA against four baselines on the **CV-Small** instance. The **Greedy Heuristic** simulates ad-hoc decision-making: it constructs a council of managers with different cost-safety weight profiles, each greedily selecting hubs and assigning demands by nearest-neighbour logic, then returns their combined non-dominated solutions. **VNS-TS** [10] is a Variable Neighbourhood Search with Tabu Search, adapted from the flood relief literature as a strong meta-heuristic baseline. **GWO-HD** [11] is a Grey Wolf Optimizer with Hamming Distance moves, representing a recent meta-heuristic designed for discrete hub-location problems. The **MILP** solver [25] applies the Adaptive Weighted Sum method [26] to systematically sweep the objective space and produces the reference Pareto front for CV-Small.

Table 1. Baseline comparison on CV-Small. HV and IGD$^+$ are normalized w.r.t. the combined non-dominated reference front across all algorithms. Best values in **bold**.

Algorithm	HV (norm.) ↑	IGD$^+$ ↓	CPU Time (s) ↓
Greedy Heuristic	0.292	0.333	$<$ **0.1**
VNS-TS	0.777	0.117	60.0
GWO-HD	0.780	0.152	0.6
MILP (Adaptive WS)	**0.997**	**0.005**	4.9
PB-NSGA (Ours)	0.925	0.039	1.1

Table 1 reveals a clear performance hierarchy. The MILP solver defines the reference Pareto front for CV-Small, establishing the quality ceiling at HV = 0.997. The Greedy heuristic runs near-instantly but remains far from the front. VNS-TS and GWO-HD both achieve competitive HV (≈0.78) with VNS-TS requiring 60 s and GWO-HD only 0.6 s. PB-NSGA closes the gap to MILP substantially (HV = 0.925, IGD$^+$ = 0.039) in 1.1 s, demonstrating that the priority-based decoder effectively navigates the combinatorial structure of MO-IHLNDP.

4.4 Experiment 2: Case Study on Central Vietnam

We apply PB-NSGA, GWO-HD, and VNS-TS to the full-scale **CV-Large** instance ($|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$), where exact solvers become intractable. All other parameters follow Sec. 4.1.

Figure 1 shows the Pareto fronts for both instances. On CV-Small (left panel), PB-NSGA closely tracks the MILP true front across the full cost-deprivation range; GWO-HD eventually converges to a similar region but with more scattered solutions; VNS-TS only finds the cost-efficient tail, missing the deprivation-minimizing end entirely. On CV-Large (right panel, power scale), the trade-off space spans nearly an order of magnitude in Z_1. PB-NSGA concentrates its solutions in the low-cost, low-deprivation region (Z_1: 4.3M–5M, Z_2: 41K–45K),

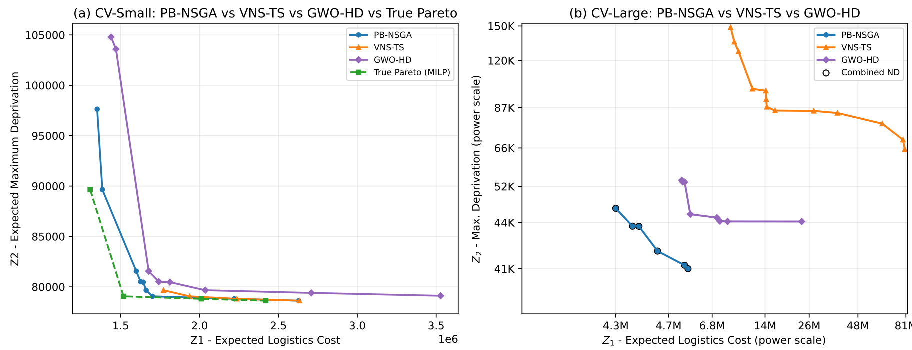

Fig. 1. Combined Pareto fronts for PB-NSGA, GWO-HD, and VNS-TS. **Left (CV-Small):** all three algorithms compared against the MILP true Pareto front. **Right (CV-Large, power scale):** combined non-dominated front across 20 runs (PB-NSGA), 5 runs (GWO-HD), and 5 runs (VNS-TS). Z_1: expected logistics cost; Z_2: expected maximum deprivation cost.

defining the efficient frontier. GWO-HD extends the front rightward, maintaining competitive $Z_2 \approx 44$K at the cost of substantially higher Z_1 (up to 26M). VNS-TS is dominated in both objectives, with solutions reaching $Z_1 = 81$M and $Z_2 = 150$K.

On CV-Large across 20 runs, PB-NSGA achieves $\text{HV} = 1.000 \pm 0.006$ and $\text{IGD}^+ = 0.000 \pm 0.001$, contributing the whole combined reference front and confirming stable convergence. GWO-HD reaches $\text{HV} = 0.977 \pm 0.008$ in 4.6 s. VNS-TS achieves $\text{HV} = 0.793 \pm 0.044$ at 180 s per run – a 20× runtime disadvantage relative to PB-NSGA.

Figure 2 visualizes the knee-point solution ($Z_1 = \$4.6$M, $Z_2 = 43{,}453$) across three scenarios. The most striking feature is structural stability: the same eight planned hubs (H0, H1, H3, H11, H14, H15, H16, H19) remain active with identical inventory fill levels across all scenarios – no reactive hub is ever opened and no lateral transshipment occurs. This reveals that the algorithm converges to a network design robust enough to serve all flood regimes purely through pre-positioning. H16 (91%) and H15 (89%) carry the heaviest pre-positioned loads, anchoring coverage for the central and southern coastal corridor. The primary adaptation across scenarios is modal: mild and severe conditions are served by a mix of truck and watercraft links, while the extreme scenario activates helicopter routes (orange dashed) to reach high-risk demand nodes that become inaccessible by road and water.

Decision Support Framework *Robust hub prioritization.* Hub selection frequency across all 20 runs on CV-Large (128 Pareto solutions total) reveals a clear investment hierarchy. The Quang Ngai Port Hub (H15) and Phuoc Son Helipad (H16) are selected in every Pareto-optimal solution (100%), making them unconditional anchors for permanent infrastructure investment. The Huong Viet

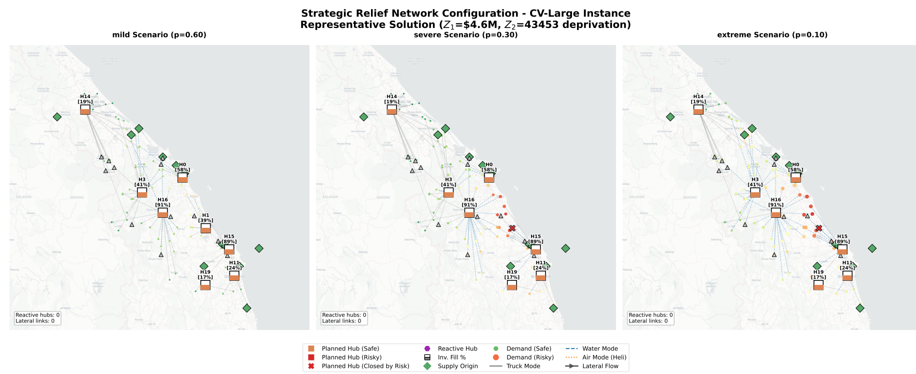

Fig. 2. Strategic relief network for a representative solution on CV-Large ($Z_1 = \$4.6$M, $Z_2 = 43{,}453$) across three probabilistic scenarios: mild ($p = 0.60$), severe ($p = 0.30$), and extreme ($p = 0.10$). Hub inventory fill levels are annotated in brackets. No reactive hub is activated in any scenario. Line styles indicate transport mode: solid gray (truck), blue dashed (water), orange dashed (helicopter).

Depot (H14, 82.8%), Da Nang Airport Hub (H0, 77.3%), Tam Ky Logistics Hub (H1, 67.2%), Binh Son Warehouse (H9, 59.4%), and Dong Giang Rescue Station (H3, 57.8%) form a second tier, each appearing in more than half of all solutions.

Resilience gap. The risk heatmap reveals a critical vulnerability: H1 (Tam Ky Logistics Hub), despite its high selection frequency, exposes a *resilience gap* – its risk escalates sharply from 0.46 under mild conditions to 0.94 and 0.92 under severe and extreme scenarios, both well above the safety threshold $\chi = 0.7$. This hub will be closed by the risk constraint whenever disaster severity escalates, requiring the network to fall back on remaining hubs. Authorities should treat H1 as a conditional asset and pre-stage contingency inventory at neighbouring safe hubs accordingly. By contrast, the Huong Viet Depot (H14) stands out as the most disaster-resilient candidate with risk values of only 0.05, 0.09, and 0.13 across all three scenarios, reflecting its sheltered inland location.

Transport mode diversification. The extreme scenario enforces a modal shift away from disrupted roads. Pre-equipping coastal hubs – anchored by Quang Ngai Port (H15) and Phuoc Son Helipad (H16) – with watercraft terminals and maintaining rotary-wing airspace access at Da Nang Airport (H0) delivers asymmetric resilience at far lower cost than deploying redundant infrastructure.

5 Conclusion and Future Work

This paper makes three contributions to humanitarian logistics under uncertainty. First, we formulate MO-IHLNDP, a two-stage stochastic bi-objective model that jointly captures incomplete hub networks, multi-modal transport, deprivation cost, and pre-positioned inventory for flood disaster relief. Second, we propose PB-NSGA, a priority-based NSGA-II with an MCF-backed decoder that efficiently navigates the coupled first- and second-stage decisions, outperforming exact and meta-heuristic baselines on verifiable instances while scaling to commune-level resolution in under 10 s. Third, applied to four flood-prone provinces in Central Vietnam, the joint framework delivers a practical decision-support tool: a quantified cost-deprivation trade-off, a stable hub investment hierarchy anchored by the Quang Ngai Port Hub and Phuoc Son Helipad, and explicit identification of resilience gaps for contingency planning.

Limitations. The model is validated on a synthetic instance calibrated from administrative and geographic data; field validation against actual disaster records is necessary before operational deployment.

Future work will pursue three directions. First, integrating field-validated data into a decision-support ecosystem capable of ingesting real-time sensor and satellite feeds to enable live operational deployment. Second, extending the model to better quantify the effect of black-swan events – explicitly penalizing increased hub processing times and costs under high-risk conditions to bridge the gap between scenario-based planning and tail-risk management. Third, scaling PB-NSGA to hour-long runtime budgets with continuous convergence guarantees, ensuring meaningful solution improvement for large-scale regional deployments.

References

1. Center for Disaster Philanthropy, "2025 southeast asia severe storms," December 2025.
2. Voice of Vietnam, "(translated) natural disasters in 2025: Fierce, abnormal, causing heavy losses in humans and property," December 2025. VOV.VN.
3. N. Altay and W. G. Green III, "Or/ms research in disaster operations management," *European journal of operational research*, vol. 175, no. 1, pp. 475–493, 2006.
4. L. N. Van Wassenhove, "Humanitarian aid logistics: supply chain management in high gear," *Journal of the Operational research Society*, vol. 57, no. 5, pp. 475–489, 2006.
5. Z. Dönmez, B. Y. Kara, Ö. Karsu, and F. Saldanha-da Gama, "Humanitarian facility location under uncertainty: Critical review and future prospects," *Omega*, vol. 102, p. 102393, 2021.
6. C. Boonmee, M. Arimura, and T. Asada, "Facility location optimization model for emergency humanitarian logistics," *International journal of disaster risk reduction*, vol. 24, pp. 485–498, 2017.
7. J. Holguín-Veras, N. Pérez, M. Jaller, L. N. Van Wassenhove, and F. Aros-Vera, "On the appropriate objective function for post-disaster humanitarian logistics models," *Journal of Operations Management*, vol. 31, no. 5, pp. 262–280, 2013.
8. J. F. Campbell, "Integer programming formulations of discrete hub location problems," *European Journal of Operational Research*, vol. 72, no. 2, pp. 387–405, 1994.
9. S. A. Alumur, B. Y. Kara, and O. E. Karasan, "The design of single allocation incomplete hub networks," *Transportation Research Part B: Methodological*, vol. 43, no. 10, pp. 936–951, 2009.
10. O. Sangsawang and S. Chanta, "Capacitated single-allocation hub location model for a flood relief distribution network," *Computational Intelligence*, vol. 37, no. 1, pp. 345–374, 2020.

11. C. Li, P. Han, M. Zhou, and M. Gu, "Design of multimodal hub-and-spoke transportation network for emergency relief under covid-19 pandemic: A meta-heuristic approach," *Applied Soft Computing*, vol. 133, p. 109925, 2023.
12. S. Tofighi, S. A. Torabi, and S. A. Mansouri, "Humanitarian logistics network design under mixed uncertainty," *European journal of operational research*, vol. 250, no. 1, pp. 239–250, 2016.
13. M. Yahyaei and A. Bozorgi-Amiri, "Robust reliable humanitarian relief network design: an integration of shelter and supply facility location," *Annals of Operations Research*, vol. 283, pp. 897–916, 2019.
14. F. S. Salman and E. Yücel, "Emergency facility location under random network damage: Insights from the istanbul case," *Computers & Operations Research*, vol. 62, pp. 266–281, 2015.
15. K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: Nsga-ii," *IEEE transactions on evolutionary computation*, vol. 6, no. 2, pp. 182–197, 2002.
16. Y. Zhou, J. Liu, Y. Zhang, and X. Gan, "A multi-objective evolutionary algorithm for multi-period dynamic emergency resource scheduling problems," *Transportation Research Part E: Logistics and Transportation Review*, vol. 99, pp. 77–95, 2017.
17. M. Gen, F. Altiparmak, and L. Lin, "A genetic algorithm for two-stage transportation problem using priority-based encoding," *OR Spectrum*, vol. 28, no. 3, pp. 337–354, 2006.
18. C. F. Daganzo, *Logistics systems analysis.* Springer, 2005.
19. P. B. Mirchandani and R. L. Francis, *Discrete Location Theory.* New York: Wiley-Interscience, 1990.
20. Y. Liang and K.-S. Leung, "Genetic algorithm with adaptive elitist-population strategies for multimodal function optimization," *Applied Soft Computing*, vol. 11, no. 2, pp. 2017–2034, 2011.
21. E. Zitzler and L. Thiele, "Multiobjective evolutionary algorithms: a comparative case study and the strength Pareto approach," *IEEE Transactions on Evolutionary Computation*, vol. 3, no. 4, pp. 257–271, 1999.
22. H. Ishibuchi, H. Masuda, Y. Tanigaki, and Y. Nojima, "Modified distance calculation in generational distance and inverted generational distance," in *Evolutionary Multi-Criterion Optimization*, vol. 9019 of *Lecture Notes in Computer Science*, pp. 110–125, Springer, 2015.
23. M. T. Nguyen, Z. Sebesvari, M. Souvignet, F. Bachofer, A. Braun, M. Garschagen, U. Schinkel, L. E. Yang, L. H. Nguyen, V. Hochschild, A. Assmann, and M. Hagenlocher, "Understanding and assessing flood risk in Vietnam: Current status, persisting gaps, and future directions," *Journal of Flood Risk Management*, vol. 14, no. 2, p. e12689, 2021.
24. A. Shapiro, "Monte carlo sampling methods," *Handbooks in operations research and management science*, vol. 10, pp. 353–425, 2003.
25. L. Perron and V. Furnon, "Or-tools," 2025.
26. I. Y. Kim and O. L. De Weck, "Adaptive weighted-sum method for bi-objective optimization: Pareto front generation," *Structural and multidisciplinary optimization*, vol. 29, no. 2, pp. 149–158, 2005.